



UNIVERSAL BEAMS AND COLUMNS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	I _x		I _y		i _x		i _y		Z _x		Z _y	
in (mm)	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
W4 4 x 4 (102 x 102)	11.3	470	3.86	161	1.72	4.37	1.00	2.54	5.46	89.5	1.90	31.1
W5 5 x 3 (127 x 76)	11.5	477	1.30	56	2.10	5.33	0.72	1.83	4.60	75	0.90	15.0
W5 5 x 5 (127 x 127)	21.4	891	7.51	313	2.13	5.41	1.26	3.20	8.55	140	3.00	49.2
	26.3	1,090	9.13	380	2.17	5.51	1.28	3.25	10.2	167	3.63	59.5
W6 6 x 3 1/2 (152 x 89)	20.1	838	2.20	90	2.52	6.40	0.83	2.10	6.70	110	1.20	20.0
W6 6 x 4 (152 x 102)	16.4	683	2.20	91.6	2.47	6.27	0.905	2.30	5.56	91.1	1.11	18.2
	22.1	920	2.99	124	2.49	6.32	0.918	2.33	7.31	120	1.50	24.6
	32.1	1,340	4.43	184	2.60	6.60	0.966	2.45	10.2	167	2.20	36.1
W6 6 x 6 (152 x 152)	29.1	1,210	9.32	388	2.56	6.50	1.45	3.68	9.72	159	3.11	51.0
	30.3	1,263	9.7	403	2.56	6.51	1.45	3.68	10.1	165.7	3.20	52.95
	41.4	1,720	13.3	554	2.66	6.76	1.50	3.81	13.4	220	4.41	72.3
	53.4	2,220	17.1	712	2.70	6.86	1.52	3.86	16.7	274	5.61	91.9
W7 7 x 4 (178 x 102)	32.6	1,357	3.30	138	2.95	7.49	0.940	2.39	9.30	153	1.60	27.0
W8 8 x 4 (203 x 102)	30.8	1,280	2.09	87	3.22	8.18	0.841	2.14	7.81	128	1.06	17.4
	39.6	1,650	2.73	114	3.21	8.15	0.843	2.14	9.91	162	1.37	22.4
	48.0	2,000	3.41	142	3.29	8.36	0.876	2.22	11.8	193	1.70	27.9
	50.2	2,091	3.90	163	3.34	8.49	0.93	2.37	12.6	206	2.00	32.0
W8 8 x 5 1/4 (203 x 133)	56.4	2,348	6.72	280	3.36	8.53	1.16	2.95	14.1	231	2.60	42.6
	61.9	2,580	7.97	332	3.43	8.71	1.23	3.12	15.2	249	3.04	49.8
	69.2	2,981	8.50	354	3.43	8.71	1.20	3.05	17.0	279	3.20	52.4
	75.3	3,130	9.77	407	3.49	8.86	1.26	3.20	18.2	298	3.71	60.8
W8 8 x 6 1/2 (203 x 165)	82.8	3,450	18.3	762	3.42	8.69	1.61	4.09	20.9	342	5.63	92.3
	98.0	4,080	21.7	903	3.45	8.76	1.62	4.11	24.3	398	6.63	109
W8 8 x 8 (203 x 203)	110	4,580	37.1	1,540	3.47	8.81	2.02	5.13	27.5	451	9.27	152
	127	5,290	42.6	1,770	3.51	8.92	2.03	5.16	31.2	511	10.6	174
	146	6,080	49.1	2,040	3.53	8.97	2.04	5.18	35.5	582	12.2	200
	184	7,660	60.9	2,530	3.61	9.17	2.08	5.28	43.3	710	15.0	246
	228	9,490	75.1	3,130	3.65	9.27	2.10	5.33	52.0	852	18.3	300
	272	11,300	88.6	3,690	3.72	9.45	2.12	5.38	60.4	990	21.4	351
W10 10 x 4 (254 x 102)	53.8	2,240	2.18	90.7	3.90	9.91	0.785	1.99	10.9	179	1.10	18.0
	68.9	2,870	2.89	120	3.95	10.0	0.81	2.06	13.8	226	1.45	23.8
	81.9	3,410	3.56	148	4.05	10.3	0.845	2.15	16.2	265	1.78	29.2
	96.3	4,010	4.29	179	4.14	10.5	0.874	2.22	18.8	308	2.14	35.1
W10 10 x 5 3/4 (254 x 146)	106	4,425	9.7	404	4.14	10.5	1.25	3.18	21.5	352	3.40	55.7
	118	4,910	11.4	474	4.27	10.8	1.33	3.38	23.2	380	3.97	65.1
	133	5,545	12.7	529	4.26	10.8	1.31	3.33	26.4	433	4.40	72.1
	144	5,990	14.1	587	4.35	11.0	1.36	3.45	27.9	457	4.89	80.1
	157	6,548	15.2	633	4.29	10.9	1.34	3.40	30.8	505	5.20	85.2
	170	7,080	16.8	699	4.38	11.1	1.38	3.51	32.5	533	5.76	94.4
W10 10 x 8 (254 x 203)	170	7,080	36.6	1,520	4.19	10.6	1.94	4.93	35.0	574	9.20	151
	209	8,700	45.0	1,870	4.27	10.8	1.98	5.03	42.1	690	11.3	185
	248	10,300	53.4	2,220	4.32	11.0	2.01	5.11	49.1	805	13.3	218